Lab 11: Synthetic Control

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Wenhao Jiang

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1 Goals

1. Construct a synthetic control and read the weight table.
2. Judge the design by pre-treatment fit, not by the size of the gap.
3. Run placebo inference in space and in time; build a placebo confidence interval.
4. Check robustness by leaving donors out.
5. Run penalized synthetic control and synthetic DiD.

```r
library(dplyr)
library(ggplot2)
library(tidysynth)
library(pensynth)   # penalized synthetic control (Abadie and L'Hour 2021)
## synthdid is on GitHub only:
## remotes::install_github("synth-inference/synthdid")
library(synthdid)

## the ADH (2010) panel: 39 states (California + 38 donors), 1970-2000
data("smoking", package = "tidysynth")
c(rows = nrow(smoking), states = n_distinct(smoking$state),
  years = paste(range(smoking$year), collapse = "-"))
#>        rows      states       years
#>      "1209"        "39" "1970-2000"
```

2 The design

Question. What did California's Proposition 99 (1988) do to cigarette consumption?

Treated unit. California. **Donor pool.** 38 states that adopted no comparable programme. **Outcome.** Per-capita cigarette sales. **Pre-period.** 1970–1988.

> **Note**
>
> The donor pool was restricted *on substantive grounds before estimating*: states with their own large tobacco taxes or control programmes are excluded, because they are treated too. That decision is part of the design, not a robustness check.

```r
## one line per state; California in red
smoking |>
  mutate(ca = state == "California") |>
  ggplot(aes(year, cigsale, group = state, colour = ca, linewidth = ca)) +
  geom_line(alpha = .8) +
  scale_colour_manual(values = c("grey78", "firebrick"), guide = "none") +
  scale_linewidth_manual(values = c(.3, 1.1), guide = "none") +
  geom_vline(xintercept = 1988, linetype = 2) +
  labs(x = NULL, y = "packs per capita") +
  theme_minimal(base_size = 10)
```

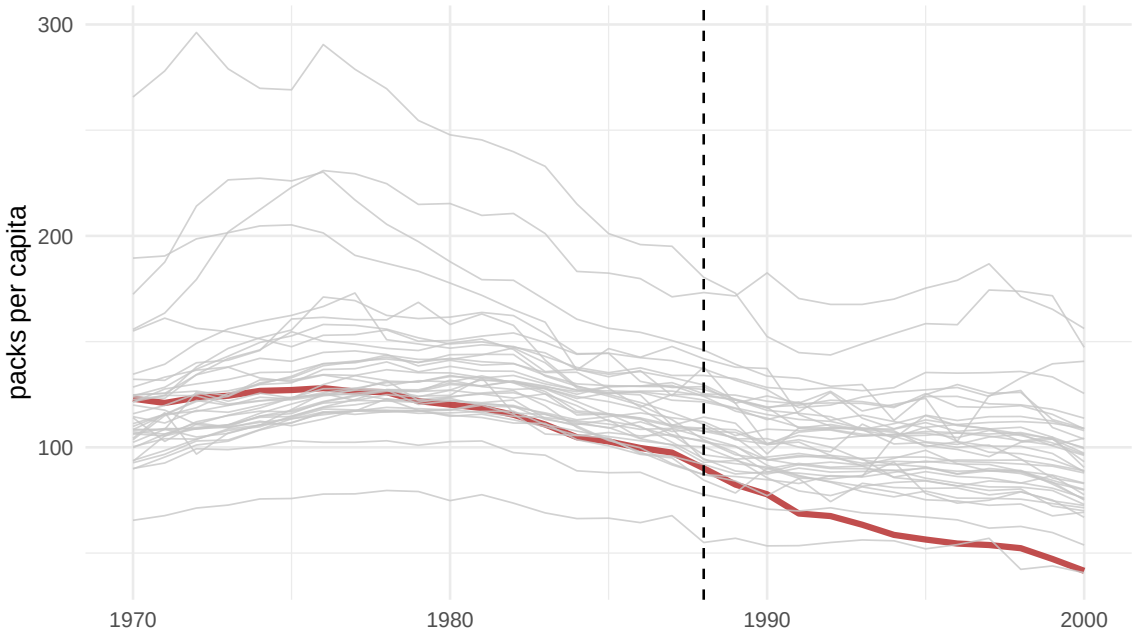


Figure 1: Raw trajectories. California is one line among 39; the eye cannot pick out a treatment effect.

3 Building the synthetic control

```
## declare outcome, unit, time, treated unit, last pre-treatment year;
## generate_placebos = TRUE also fits a synthetic control for every donor
sc <- smoking |>
  synthetic_control(outcome = cigsale, unit = state, time = year,
                    i_unit = "California", i_time = 1988,
                    generate_placebos = TRUE) |>
  ## covariates averaged over the pre-period (the v step weights these)
  generate_predictor(time_window = 1980:1988,
                     ln_income = mean(lnincome, na.rm = TRUE),
                     ret_price = mean(retprice, na.rm = TRUE),
                     youth     = mean(age15to24, na.rm = TRUE)) |>
  generate_predictor(time_window = 1984:1988,
                     beer_sales = mean(beer, na.rm = TRUE)) |>
  ## ADH also use three pre-period years of the outcome as predictors
  generate_predictor(time_window = 1975, cigsale_1975 = cigsale) |>
  generate_predictor(time_window = 1980, cigsale_1980 = cigsale) |>
  generate_predictor(time_window = 1988, cigsale_1988 = cigsale) |>
  ## outer step: choose v so that w(v) fits the 1970-88 outcomes best
  ## (optimizer settings as in the lecture script figs/w11_prop99.R)
  generate_weights(optimization_window = 1970:1988,
                   margin_ipop = .02, sigf_ipop = 7, bound_ipop = 6) |>
```

```
## build synthetic California for every year with the chosen w
generate_control()
```

> **Note**
>
> **Two kinds of years.** Each predictor is *one number per state*: the mean of price, income, and the youth share over 1980–88, mean beer sales over 1984–88 (beer is only recorded from 1984), and sales in 1975, 1980, and 1988. The inner step matches these numbers. The outer step, `optimization_window = 1970:1988`, then fits sales in *every* pre-year. Averaging the covariates is a choice: they move slowly, so a mean captures each state's level, and matching every year of four covariates would mean 76 predictors for 38 donors. The 1980–88 window is ADH's choice, not a rule; exercise 2 asks how much it matters.

3.1 Who is synthetic California?

```
## w: the donor weights (they are >= 0 and sum to 1)
sc |> grab_unit_weights() |> filter(weight > 0.001) |>
  arrange(desc(weight)) |> knitr::kable(digits = 3)
```

unit	weight
Utah	0.342
Nevada	0.238
Montana	0.209
Colorado	0.149
Connecticut	0.062

```
## v: the predictor weights (a tuning knob, not importance)
sc |> grab_predictor_weights() |> arrange(desc(weight)) |> knitr::kable(digits
= 3)
```

variable	weight
cigsale_1975	0.468
cigsale_1980	0.412
ret_price	0.055
cigsale_1988	0.037
beer_sales	0.020
youth	0.007
ln_income	0.000

> **!** Important
>
> **Print the weight table.** It is the single most informative output of a synthetic control study: readers can decide for themselves whether a weighted average of these states is a plausible California. No regression coefficient offers that.

3.2 Balance

```
## each predictor: California, synthetic California, plain donor average
sc |> grab_balance_table() |> knitr::kable(digits = 2)
```

variable	California	synthetic_California	donor_sample
ln_income	10.08	9.85	9.83
ret_price	89.42	89.39	87.27
youth	0.17	0.17	0.17
beer_sales	24.28	24.22	23.66
cigsale_1975	127.10	126.99	136.93
cigsale_1980	120.20	120.22	138.09
cigsale_1988	90.10	91.39	113.82

4 The result

```
## actual vs synthetic California, all years
plot_trends(sc)
```

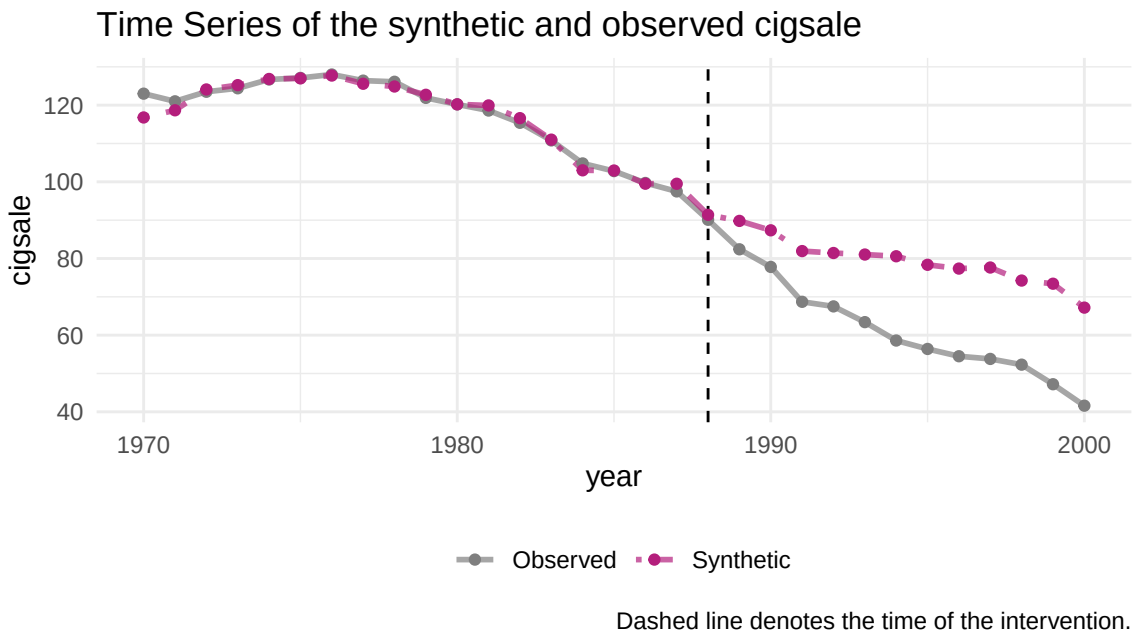


Figure 2: California and synthetic California. The pre-1988 lines nearly coincide, which is what licenses reading the post-1988 gap as an effect.

```
## the gap: actual minus synthetic
plot_differences(sc)
```

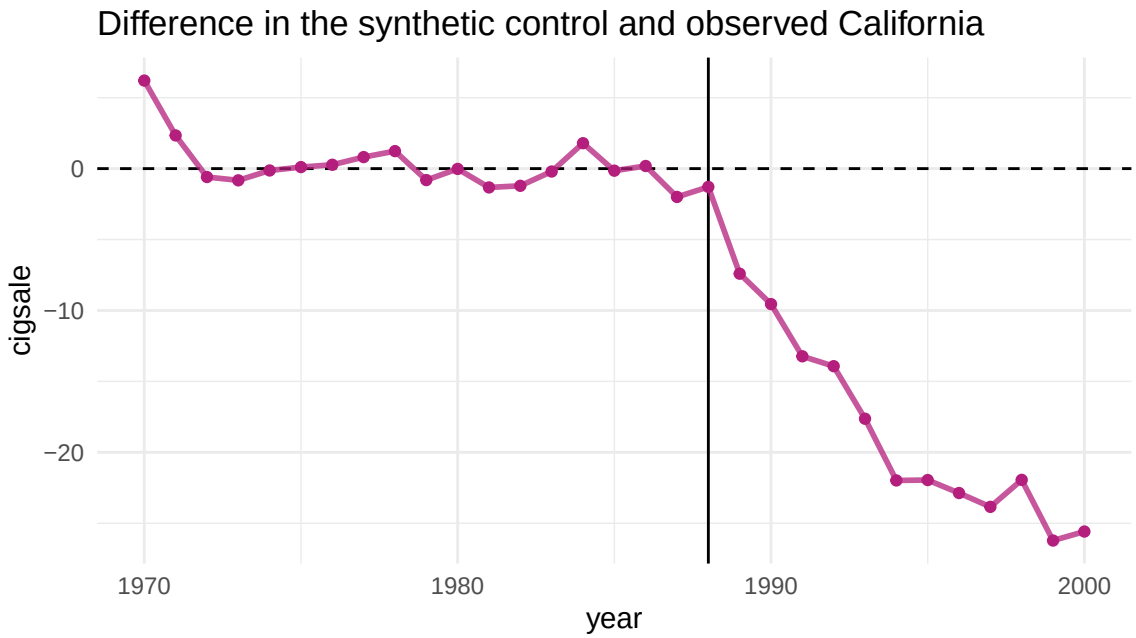


Figure 3: The gap between California and its synthetic control.

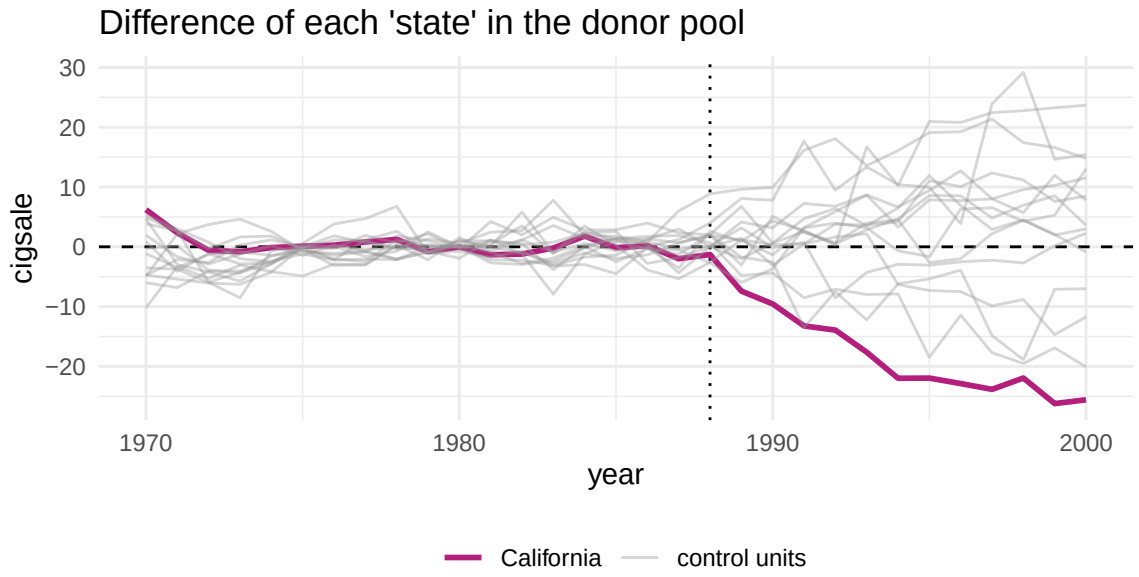
```r
## the year-by-year gap; the effect is its average after 1988
gaps <- sc |> grab_synthetic_control() |>
  mutate(gap = real_y - synth_y)
c(mean_gap_post = mean(gaps$gap[gaps$time_unit > 1988]),
  gap_2000 = gaps$gap[gaps$time_unit == 2000],
  rmspe_pre = sqrt(mean(gaps$gap[gaps$time_unit <= 1988]^2)))
#> mean_gap_post      gap_2000     rmspe_pre
#>       -18.846       -25.585         1.779
```

The pre-treatment RMSPE is the number to look at first. If it were large relative to the post-treatment gap, there would be nothing to report.

5 Inference

5.1 Placebos in space

```r
## every donor's gap when it is treated as the treated unit
plot_placebos(sc)
```



ıed all placebo cases with a pre−period RMSPE exceeding two times the treated unit's pre−period RMSPE.

Figure 4: Each grey line is a donor state treated as if it had adopted Proposition 99. California is in black.

`plot_placebos()` drops donors whose pre-treatment fit is much worse than California's — otherwise states the method could never match would dominate the picture for reasons unrelated to treatment.

5.2 The RMSPE ratio

```
## post/pre ratio: a big post gap counts only if the pre fit was good
plot_mspe_ratio(sc)
```

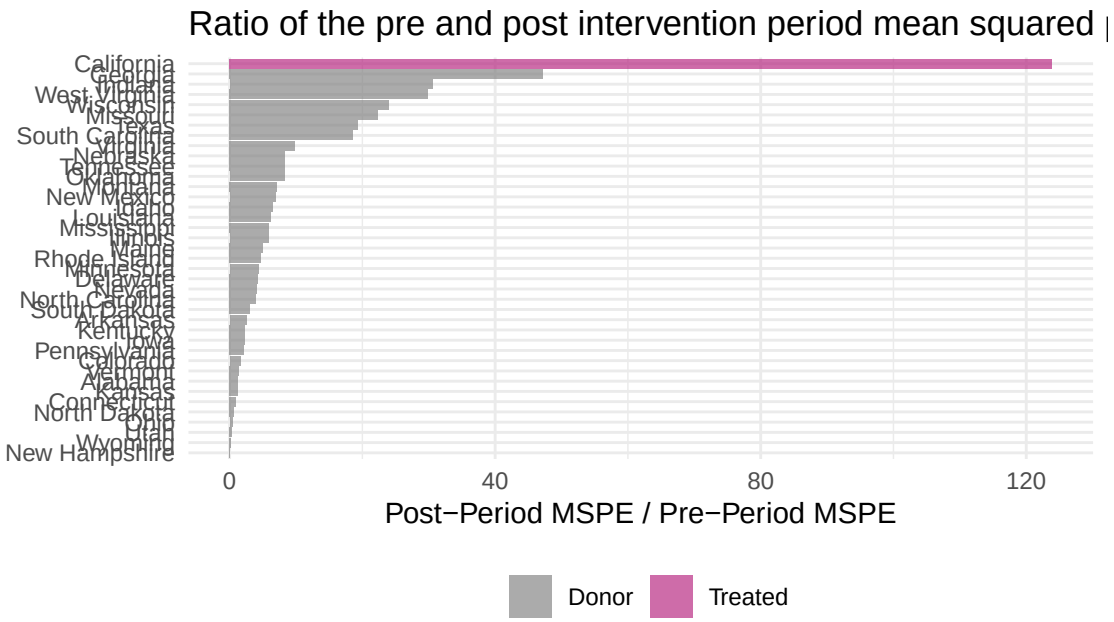


Figure 5: Ratio of post- to pre-treatment RMSPE. California is the largest of all 39 units.

```
## rank of each unit by its ratio; p = rank / 39
sig <- sc |> grab_significance()
sig |> select(unit_name, type, mspe_ratio, rank, fishers_exact_pvalue) |>
  head(6) |> knitr::kable(digits = 3)
```

unit_name	type	mspe_ratio	rank	fishers_exact_pvalue
California	Treated	123.87	1	0.026
Georgia	Donor	47.21	2	0.051
Indiana	Donor	30.58	3	0.077
West Virginia	Donor	29.83	4	0.103
Wisconsin	Donor	24.05	5	0.128
Missouri	Donor	22.38	6	0.154

The `fishers_exact_pvalue` column is the unit's rank divided by the number of units. This is a permutation test, and it is what you report as the p-value: if the law did nothing and California were one of 39 interchangeable states, a ratio this large would come up 1 time in 39. With 38 donors the smallest attainable value is $1/39 \approx 0.026$, a floor worth stating when you report it. It is exact only if the treated state had been picked at random.

5.3 A confidence interval from the placebos

The rank gives a p-value but no interval. A second option (lecture: *A Standard Error From the Placebos*): treat the placebo states' "effects" as draws of what the estimator produces when nothing happened, and use their standard deviation as the standard error.

```
## every unit's fit (California and the 38 placebo states), all years
pl <- sc |> grab_synthetic_control(placebo = TRUE)

## each unit's "effect": its average gap after 1988
eff <- pl |> filter(time_unit > 1988) |>
  group_by(.id, .placebo) |>
  summarise(effect = mean(real_y - synth_y), .groups = "drop")
est <- eff$effect[eff$.placebo == 0]          # California
se  <- sd(eff$effect[eff$.placebo == 1])      # spread of the 38 placebos
round(c(estimate = est, se = se, lo = est - 1.96 * se, hi = est + 1.96 * se),
1)
#> estimate       se       lo       hi
#>    -18.8     13.8    -46.0      8.3
```

Placebo states the method fits badly have large "effects" for reasons that have nothing to do with a law, and they inflate the standard error. Keep only the placebos that fit about as well as California (pre-period RMSPE at most twice California's, the rule `plot_placebos()` uses):

```
## pre-period RMSPE for every unit
fit_pre <- pl |> filter(time_unit <= 1988) |>
  group_by(.id) |> summarise(rmspe = sqrt(mean((real_y - synth_y)^2)))
ca_rmspe <- fit_pre$rmspe[fit_pre$.id == "California"]
good <- fit_pre$.id[fit_pre$rmspe <= 2 * ca_rmspe & fit_pre$.id !=
"California"]

## the same interval from the well-fitting placebos only
se_good <- sd(eff$effect[eff$.id %in% good])
round(c(placebos = length(good), se = se_good,
        lo = est - 1.96 * se_good, hi = est + 1.96 * se_good), 1)
#> placebos       se       lo       hi
#>     14.0      8.3    -35.1     -2.6
```

With all 38 placebos the interval includes 0; with the 14 that fit well it does not. Say which placebos you used. The rank p-value (0.026) does this discounting automatically, through the RMSPE ratio.

5.4 A placebo in time

Pretend the programme began in 1981, eight years early, and refit using only 1970–1988. ADH's predictors are moved back to the fake date: covariates averaged over 1972–80 (income starts in 1972; beer is not recorded yet) and sales in 1970, 1975, and 1980. Nothing should happen in 1981–88.

```
## keep only pre-law years; the last "pre" year is now 1980
sc_1980 <- smoking |>
  filter(year <= 1988) |>
  synthetic_control(outcome = cigsale, unit = state, time = year,
                    i_unit = "California", i_time = 1980,
                    generate_placebos = FALSE) |>
  generate_predictor(time_window = 1972:1980,
                     ln_income = mean(lnincome, na.rm = TRUE),
                     ret_price = mean(retprice, na.rm = TRUE),
                     youth     = mean(age15to24, na.rm = TRUE)) |>
  generate_predictor(time_window = 1970, cigsale_1970 = cigsale) |>
  generate_predictor(time_window = 1975, cigsale_1975 = cigsale) |>
  generate_predictor(time_window = 1980, cigsale_1980 = cigsale) |>
  generate_weights(optimization_window = 1970:1980,
                   margin_ipop = .02, sigf_ipop = 7, bound_ipop = 6) |>
  generate_control()

## the fake effect: average gap 1981-88 (should be near 0)
g80 <- sc_1980 |> grab_synthetic_control() |> mutate(gap = real_y - synth_y)
c(mean_gap_1981_1988 = mean(g80$gap[g80$time_unit > 1980]),
  rmspe_pre = sqrt(mean(g80$gap[g80$time_unit <= 1980]^2)))
#> mean_gap_1981_1988          rmspe_pre
#>            -2.8662             0.8942
```

The fake gap is about −2.9 packs, roughly 15% of the real effect: California was already pulling slightly away in the 1980s. Report it. A fake gap as large as the real one would mean the method manufactures effects.

5.5 Leave-one-donor-out

```
## donors that carry weight
donors <- sc |> grab_unit_weights() |> filter(weight > 0.01) |> pull(unit)

## refit the EXACT original pipeline after dropping each donor, so that any
## change is attributable to the donor and not to a specification change
loo <- lapply(donors, function(dd) {
  s <- smoking |> filter(state != dd) |>
```

```
    synthetic_control(outcome = cigsale, unit = state, time = year,
                      i_unit = "California", i_time = 1988,
                      generate_placebos = FALSE) |>
    generate_predictor(time_window = 1980:1988,
                       ln_income = mean(lnincome, na.rm = TRUE),
                       ret_price = mean(retprice, na.rm = TRUE),
                       youth     = mean(age15to24, na.rm = TRUE)) |>
    generate_predictor(time_window = 1984:1988,
                       beer_sales = mean(beer, na.rm = TRUE)) |>
    generate_predictor(time_window = 1975, cigsale_1975 = cigsale) |>
    generate_predictor(time_window = 1980, cigsale_1980 = cigsale) |>
    generate_predictor(time_window = 1988, cigsale_1988 = cigsale) |>
    generate_weights(optimization_window = 1970:1988,
                     margin_ipop = .02, sigf_ipop = 7, bound_ipop = 6) |>
    generate_control()
  ## the estimate without donor dd
  g <- s |> grab_synthetic_control() |> mutate(gap = real_y - synth_y)
  tibble::tibble(dropped = dd, mean_gap = mean(g$gap[g$time_unit > 1988]))
}) |> bind_rows()

knitr::kable(loo, digits = 2)
```

dropped	mean_gap
Colorado	-19.28
Connecticut	-18.70
Montana	-17.63
Nevada	-20.62
Utah	-18.16

If dropping any single donor moved the estimate substantially, the result would rest on one state rather than on a comparison group.

6 Penalized synthetic control

`pensynth` implements the penalized estimator from lecture: it adds $\lambda \sum_j w_j \sum_k (X_{1k} - X_{jk})^2$, the distance to each donor used, to the usual predictor fit. `lambda = 0` is plain synthetic control.

```
## the lecture's toy: one predictor, California = 5, donors A 4, B 6, C 0, D
10
## X1: the treated unit's predictors (K x 1); X0: the donors' (K x J)
quiet <- clarabel::clarabel_control(verbose = FALSE)   # silence the solver
```

```r
toy <- pensynth(X1 = matrix(5), X0 = matrix(c(4, 6, 0, 10), nrow = 1),
                lambda = 0.1, standardize = FALSE, verbose = FALSE,
                opt_pars = quiet)
round(toy$w, 3)   ## all weight on the nearby donors A and B
#> [1] 0.5 0.5 0.0 0.0
```

On Prop 99 we reuse the predictors and the predictor weights v from the `tidysynth` fit above, so that `lambda = 0` reproduces synthetic California.

```r
## predictors: one column per state (California; the 38 donors)
X1 <- as.matrix(grab_predictors(sc, type = "treated")[, -1])
X0 <- as.matrix(grab_predictors(sc, type = "controls")[, -1])
v  <- grab_predictor_weights(sc)$weight     # the v chosen by tidysynth

## outcomes as a state x year matrix, for the gaps
Y <- smoking |> select(state, year, cigsale) |>
  tidyr::pivot_wider(names_from = year, values_from = cigsale)
Y <- as.matrix(tibble::column_to_rownames(Y, "state"))
pre <- as.character(1970:1988); post <- as.character(1989:2000)

## fit for several lambdas; report donors used, average gap, pre-period RMSPE
pen_row <- function(lam) {
  w <- pensynth(X1, X0, v = v, lambda = lam, verbose = FALSE,
                opt_pars = quiet)$w
  g <- Y["California", ] - colSums(w * Y[colnames(X0), ])
  top <- sort(setNames(w, colnames(X0)), decreasing = TRUE)[1:2]
  tibble::tibble(lambda = lam, donors = sum(w > 0.005),
                 gap_post = round(mean(g[post]), 1),
                 rmspe_pre = round(sqrt(mean(g[pre]^2)), 2),
                 top_two = paste(names(top), round(top, 2), collapse = ", "))
}
lapply(c(0, 0.001, 0.01, 0.1, 1), pen_row) |> bind_rows() |>
  knitr::kable()
```

lambda	donors	gap_post	rmspe_pre	top_two
0.000	5	-18.8	1.78	Utah 0.34, Nevada 0.24
0.001	6	-18.8	2.39	Utah 0.27, Nevada 0.22
0.010	5	-27.2	7.08	North Dakota 0.39, Idaho 0.28
0.100	3	-29.3	6.67	Montana 0.41, North Dakota 0.32
1.000	3	-31.5	7.44	Iowa 0.78, Montana 0.21

Larger λ pulls the weights toward the states closest to California on the predictors, and the

pre-period fit gets worse. `cv_pensynth()` picks λ by how well the weights fit the pre-period *outcomes* (`Z1`, `Z0`):

```
## Z1, Z0: California's and the donors' sales in 1970-88 (years in rows)
cv <- cv_pensynth(X1, X0, Z1 = matrix(Y["California", pre]),
                  Z0 = t(Y[colnames(X0), pre]), v = v,
                  verbose = FALSE, opt_pars = quiet)
signif(cv$l_opt, 2)   ## the chosen lambda
#> [1] 0.00011
w_cv <- setNames(cv$w_opt, colnames(X0))
round(sort(w_cv[w_cv > 0.005], decreasing = TRUE), 3)
#>        Utah      Nevada     Montana    Colorado Connecticut
#>       0.339       0.236       0.209       0.152       0.063
```

The chosen λ is close to 0 and the weights are those of synthetic California: here the classic fit already uses nearby donors, so the penalty has nothing to fix. It matters when many different mixes fit equally well, as in the toy, or when there are many treated units, each needing its own weights.

7 Synthetic DiD

`synthdid` (Arkhangelsky et al. 2021) needs a balanced panel with a 0/1 treatment column. `panel.matrices()` turns it into an outcome matrix with the treated unit last.

```
## treated = 1 for California from 1989 on
panel <- smoking |>
  transmute(state, year, cigsale,
            treated = as.integer(state == "California" & year >= 1989)) |>
  as.data.frame()
setup <- panel.matrices(panel)      # Y (39 x 31), N0 = 38 donors, T0 = 19
years

## the three estimators on the same matrix
fits <- list(DiD  = did_estimate(setup$Y, setup$N0, setup$T0),
             SC   = sc_estimate(setup$Y, setup$N0, setup$T0),
             SDiD = synthdid_estimate(setup$Y, setup$N0, setup$T0))

## placebo standard errors: refit with a donor as the "treated" unit, many
## times, and take the sd of those estimates (random draws, so set a seed)
set.seed(723)
se_p <- sapply(fits, function(f) sqrt(vcov(f, method = "placebo")))
est_p <- sapply(fits, c)
round(cbind(estimate = est_p, se = se_p,
            lo = est_p - 1.96 * se_p, hi = est_p + 1.96 * se_p), 1)
```

```
#>       estimate   se    lo  hi
#> DiD     -27.3 16.0 -58.7 4.0
#> SC      -19.6 11.1 -41.4 2.1
#> SDiD    -15.6  8.9 -33.1 1.9
```

These are the numbers on the lecture slides. SDiD's two sets of weights:

```
w_sdid <- attr(fits$SDiD, "weights")
## donor weights omega (fitted with a free constant, so levels need not match)
omega <- setNames(w_sdid$omega, rownames(setup$Y)[1:setup$N0])
round(head(sort(omega, decreasing = TRUE), 5), 3)
#>        Nevada New Hampshire   Connecticut      Delaware      Colorado
#>         0.124         0.105         0.078         0.070         0.058
## pre-year weights lambda: which years form the "before" period
lambda <- setNames(w_sdid$lambda, colnames(setup$Y)[1:setup$T0])
round(lambda[lambda > 0.01], 3)
#>  1986  1987  1988
#> 0.366 0.206 0.427
```

```
## synthdid's own plot of the weighted 2x2
synthdid_plot(fits$SDiD) + theme_minimal(base_size = 10)
```

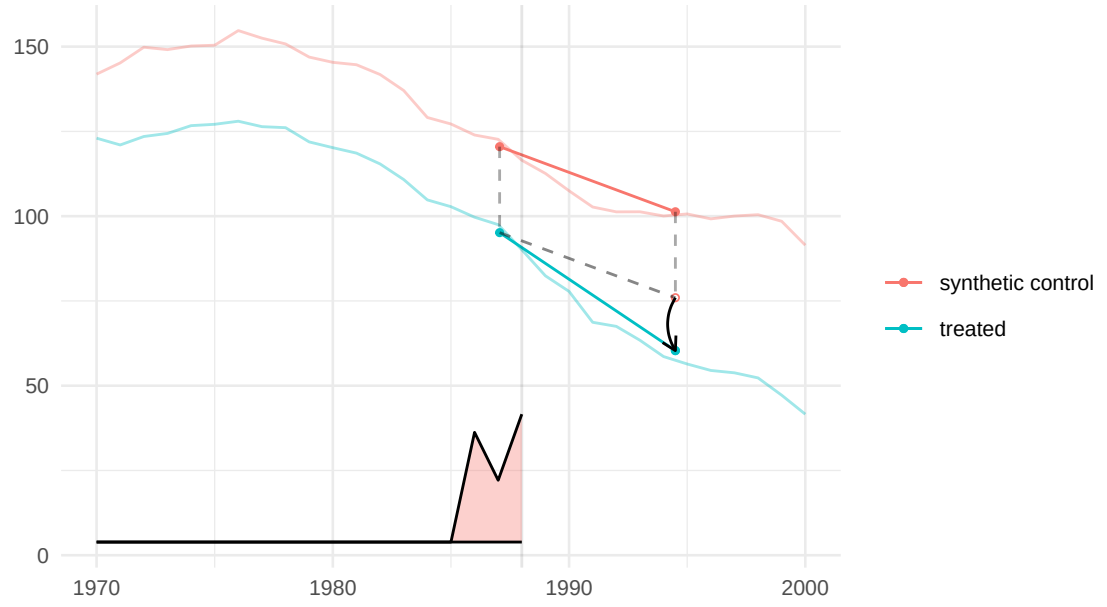


Figure 6: Synthetic DiD: California and its weighted control (parallel, not equal, before 1989). The shaded area at the bottom shows the pre-year weights (1986-88).

8 Exercises

1. **Predictor sensitivity.** Re-estimate using only lagged outcomes as predictors (drop income, price, beer, youth). How much do the weights and the estimate move? What does that tell you about how much work the covariates are doing?
2. **Windows.** (a) Change the covariate window from 1980–88 to 1972–88 (the years income is recorded). Do the weights or the estimate move? (b) Re-estimate using only 1980–1988 to fit the weights. Does the pre-treatment fit worsen? Does the estimate change? Connect this to "the fit was luck" on the lecture slide *When Parallel Trends Fails After Weighting*.
3. **Spillovers.** Nevada borders California and carries about a quarter of the weight. If Californians crossed the border to buy cigarettes, would the estimate be too large or too small? Use the leave-one-out table to see what happens without Nevada, and explain why that check cannot rule out spillovers into the other donors.
4. **DiD as a 2×2.** Compute the 2×2 comparing California with the plain average of the 38 donors, using the means for 1970–88 and 1989–2000 (lecture: −27.3). Redo it with single years (1980 → 2000, 1988 → 1995). Why does the answer depend on the years chosen?
5. **A different treated unit.** Using `set.seed(723)`, pick a donor state at random, treat it as treated, and run the whole pipeline. Write the paragraph you would write if this were your result — then explain what stops you from believing it.

9 Session information

```
sessionInfo()
#> R version 4.4.1 (2024-06-14)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS 26.5.2
#>
#> Matrix products: default
#> BLAS:
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib

#> LAPACK:
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;
LAPACK version 3.12.0
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: America/New_York
#> tzcode source: internal
```

```
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] synthdid_0.0.9  pensynth_0.8.1  tidysynth_0.2.1 ggplot2_4.0.0
#> [5] dplyr_1.1.4
#>
#> loaded via a namespace (and not attached):
#>  [1] Matrix_1.7-0        gtable_0.3.6        jsonlite_1.8.9
#>  [4] compiler_4.4.1      tidyselect_1.2.1    tinytex_0.53
#>  [7] tidyr_1.3.1         scales_1.4.0        clarabel_0.11.2
#> [10] yaml_2.3.10         fastmap_1.2.0       lattice_0.22-7
#> [13] R6_2.5.1            labeling_0.4.3      generics_0.1.3
#> [16] kernlab_0.9-33      knitr_1.48          forcats_1.0.0
#> [19] tibble_3.2.1        nloptr_2.1.1        pillar_1.9.0
#> [22] RColorBrewer_1.1-3  rlang_1.3.0         utf8_1.2.4
#> [25] xfun_0.47           S7_0.2.0            cli_3.6.5
#> [28] withr_3.0.1         magrittr_2.0.3      digest_0.6.37
#> [31] grid_4.4.1          mvtnorm_1.3-1       lifecycle_1.0.4
#> [34] vctrs_0.6.5         pracma_2.4.4        evaluate_1.0.0
#> [37] glue_1.7.0          numDeriv_2016.8-1.1 farver_2.1.2
#> [40] optimx_2024-12.2    fansi_1.0.6         rmarkdown_2.28
#> [43] purrr_1.0.2         tools_4.4.1         pkgconfig_2.0.3
#> [46] htmltools_0.5.8.1
```